

NAT and PAT -- Guided Notes

Unit 2.3.2 | CompTIA Network+ N10-009 Obj. 2.3 | N10-008 2.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

NAT stands for _____. It was developed to extend the life of IPv4 by allowing _____ of devices to share a single _____ IP address, delaying the exhaustion of the public address pool.

Question 2

The three private (non-routable) IPv4 address ranges are _____, _____, and _____. Packets with these source addresses are _____ by internet backbone routers.

Question 3

Static NAT creates a _____ -to- _____ permanent mapping between a private address and a public address. This is typically used for devices that must be _____ from the internet, such as web servers or mail servers.

Question 4

PAT (Port Address Translation) is also called NAT _____. It allows thousands of devices to share a _____ public IP address by using unique _____ numbers to track each session in the NAT translation table.

Question 5

In NAT terminology, the _____ interface faces the private network, while the _____ interface faces the public internet. When a packet leaves the private network, NAT replaces the _____ address with the router's public IP.

Question 6

Describe one advantage and one disadvantage of NAT. How does NAT affect end-to-end connectivity, and why is this a problem for certain protocols or applications?

Question 7

Dynamic NAT maps private addresses to a _____ of public addresses. Unlike PAT, each active session uses a _____ public IP. If the pool is exhausted, new connections are _____ until a public address becomes available.

Question 8 -- Real World Application

REAL WORLD: A user reports they can access the internet normally, but a colleague trying to connect to the user's PC from outside the office cannot reach it. The office uses PAT on a single public IP. Explain why this happens and describe two solutions that would allow inbound connections.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-3-2-nat-and-pat/2-3-2-nat-and-pat-answer-key.pdf
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